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Review

Spherical 3D fractal structured dual-mode triboelectric nanogenerator for multidirectional low-frequency wave energy harvesting

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ABSTRACT

The triboelectric nanogenerator (TENG) has demonstrated an irreplaceable advantage compared with the electromagnetic generator (EMG) in featuring a greater low-frequency energy harvesting capability. However, the efficiency of energy harvesting remains challenge for wave energy under random-direction motion characteristics. In this work, a 3D fractal structure dual-mode spherical TENG (FDS-TENG) is constructed for efficient wave energy harvesting. Inspired by the structure of the burr puzzle, a three-module multiphase coupling strategy is introduced to the TENG through the ingenious 3D spatial geometric isomorphism design. The synergistic coupling motion between the modules achieves a dual-mode output in contact-separation mode and lateral sliding mode. The designed structure greatly increases the space utilization of the spherical TENG while enabling efficient capture of wave energy at low frequency, weak excitation, and in any direction of activation. The peak power density and average power density of the FDS-TENG can reach 8.3 W/m^3 and 1.1 W/m^3 at 2 Hz, respectively. Based on the FDS-TENG, several applications for the smart ocean, including the power supply of beacon lights and environmental sensors, have been successfully demonstrated. This novel design strategy is a significant inspiration for improving the response characteristics and output performance of TENGs in the future.

1. Introduction

Blue energy, one of the most promising renewable and clean energy sources[1,2], has been further developed and applied owing to its advantages of widespread distribution, large energy storage, and high energy density[3,4]. Wave energy, as the most typical form of ocean energy, occurs from the kinetic energy of the wind transferred to the surface of the ocean[5], where the complexity of the ocean environment causes itself highly random, low-frequency wave characteristics in space and time, enabling a challengeable energy harvesting[6,7]. In general, wave energy harvesting mainly relies on electromagnetic generators (EMGs), in which the intrinsic structure dominated by magnets and metal coils reduces flexibility and requires large space[8]. Besides, the output power of EMGs is proportional to the squared frequency of mechanical motion, thus the EMG-based wave energy harvesting device with low output can hardly reach the working threshold of the electronic ocean device in low-frequency environments[9]. The emerging triboelectric nanogenerator (TENG) proposed by Z. L. Wang has pioneered a new path for low-frequency and disordered energy harvesting[10–12]. Based on the Maxwell's principle of displacement current, this technology operates with coupled triboelectrification and electrostatic induction that can directly convert mechanical energy into electrical energy[13]. Owing to the features of light weight, easily fabricated, and high conversion efficiency, the TENG technology has been widely applied in this field for wave energy harvesting and also further self-powered sensing[14–17].

Previously, the wave energy harvesting efficiency including the kinetic property as well as the output power has been enhanced by optimized structural designs of TENGs[18–23]. In particular, the rolling spherical structure has become a common solution for TENG design due to its sensitive rolling response characteristics, which can enable the wave energy conversion at low frequencies, weak excitations, and

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multiple directions[24,25]. However, the inherent working mode of point-to-point contact is not conducive to efficient charge transfer resulting in suboptimal output performance. In this regard, the quest for enhancing the effective contact area utilization has led to the conception of several novel spherical TENG designs. Accordingly, the elastic connections with soft contact designs provide superior durability and high efficiency[26,27]. An innovative rolling sphere structure adopts a soft-contact flexible ball design to increase the effective contact area with higher output performance over the traditional hard contact structure[28]. Better effective contact area is also achieved by using a multi-sphere design combined with 3D electrode structure to boost charge and power output by increasing the contact interface[29]. However, the effective contact area in working is still limited to a narrow surface area resulting in decreased efficiency of the overall device. Besides, the multilayer structure of TENG based on the contact-separation mode facilitates a more space utilization and improves efficient triboelectrification[30–33]. Compared to the spherical TENG, the multi-unit integrated TENG is hard to work synchronously or even activate in the face of randomly oriented waves resulting in a mismatch with the actual demand. Therefore, it is an appealing solution to improve the wave energy collection capacity by aiming at further increasing the space utilization of the spherical TENG. Inspired by the burr puzzle with a 3D symmetric coupling structure, the new 3D fractal strategy enables a multidimensional segmentation of the sphere to produce multiple identical sub-modules that are coupled to each other, thus generating a larger contact area inside. Through rational structural design, this structure has the potential to greatly maximize the space utilization of the TENG and realize the synchronous and cooperative work of each unit for efficient multi-directional wave energy harvesting over a wide frequency range.

In this work, a 3D fractal structure dual-mode spherical TENG (FDS-TENG) is proposed to adapt wave energy harvesting. Based on the burr-puzzle structures, a three-module multiphase coupling strategy is introduced through a 3D spatial geometrical isomorphic design, which greatly increases the space utilization of the spherical TENG. By cutting the sphere into three identical coupled component units with complementary spatial symmetry, the model achieves coupled motion between the units. The triboelectric dielectric layers with different dimensions is formed in the inner units, which synergistically realizes the dual-mode output of the contact-separation mode and the lateral sliding mode. The experimental results show that the fabricated FDS-TENG maintains a favorable energy conversion efficiency in the wave frequency range of 0.5–3 Hz through the effective complementarity of different operating modes. Through structural parameters optimization, the maximum contact area and optimal response characteristics can be well matched. The FDS-TENG can achieve wave energy harvesting in any direction within the range of 0.5–3 Hz. with short-circuit current up to 18 μA and open-circuit voltage up to 820 V at 2 Hz. Finally, the potential of TENG application in smart ocean is demonstrated by applying the FDS-TENG to temperature and humidity sensors and beacon lights.

2. Results and discussion

2.1. Structure and working mechanism of the FDS-TENG

Fig. 1(a) envisions the future ocean system with new type of distributed energy that supply ocean monitoring devices, such as ocean distributed sensors, weather sensors and marine beacon lights by the proposed FDS-TENG. The overall structure of the FDS-TENG is shown in Fig. 1(b), which mainly consists of a sphere made of resin material and an acrylic encapsulated spherical shell. The sphere is divided into three identical coupled component units by designing its 3D spatial geometry, which are embedded with each other to form burr-puzzle structures, as the physical image shown in Fig. S1. For a single component unit, foam adhesive is attached as a cushioning layer to all cutting surfaces on both sides of the interior, copper foils are attached as electrodes on top of the cushioning layer, and a flexible polytetrafluoroethylene (PTFE) film completely covers one of the sides as an electronegative dielectric

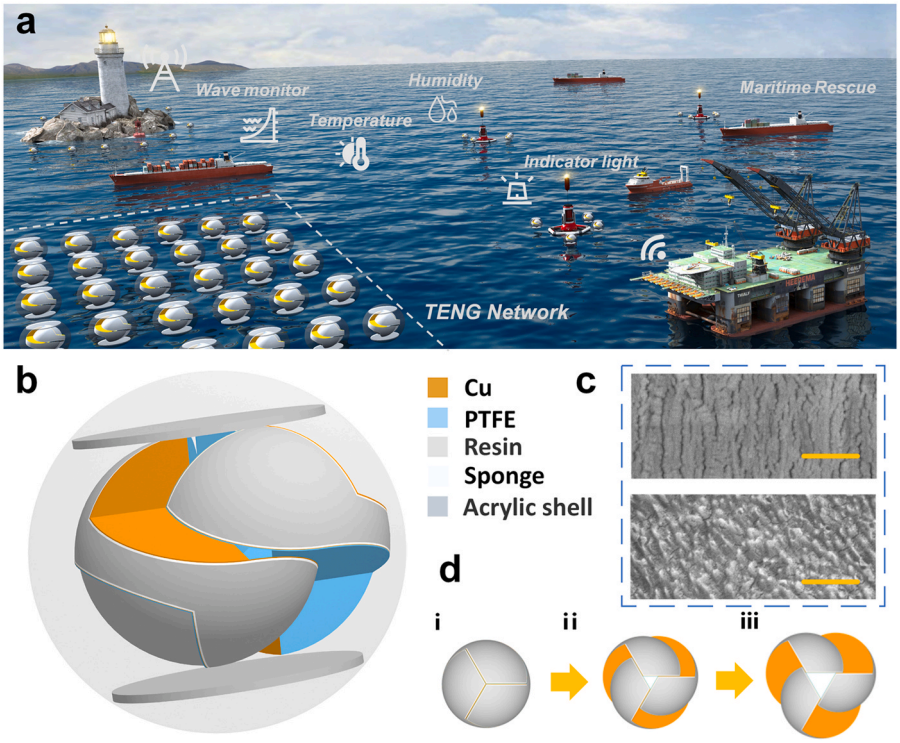


Fig. 1. Structural design of a dual-mode spherical TENG based on an efficient fractal strategy. a) Schematic diagram of the TENG network for wave energy harvesting and ocean sensing. b) Structural composition of the TENG. c) SEM images of the PTFE surfaces and after nanostructured, with a scale of 1 μm. d) Working principle of the TENG coupled motion.

material. The surface of the PTFE film was modified by plasma-etching method to create nanostructures with larger contacts, which are more hydrophobic to reduce the moisture interference in the marine environment, as shown in the scanning electron microscope (SEM) image in Fig. 1(c). The two-part units are cross-coupled with each other, so that all internal cutting surfaces form triboelectric dielectric layers between the copper-coated PTFE film and the copper foil. Fig. 1(d) shows the motion trajectory of the FDS-TENG in the overhead view. When the FDS-TENG is subjected to the wave-excited inertial force and the gravity force generated by tilting, the sub-module with the force direction similar to the displacement direction will slide outward and drive the other coupling components to conduct the separation motion with a circumferential interval of 120° as shown in Fig. S2. In the other case, when the tilting direction is different from the displacement direction of the component units, the component unit fixed to the rotation axis will rotate the sphere to self-adapt the wave direction for the separation motion. In this process, the three units can effectively realize the concerted motion of opening-closing under one wave cycle, in which the internal cut surface between each of two units works as a contact surface. In order to prevent the units from failing due to separation, a reasonably sized encapsulated spherical shell will limit the maximum displacement of the internal sub-units. After a certain distance of separation, the three units can still slide inward under the action of external force to realize the amalgamation, where separation and amalgamation are interlinked and mutually reinforced. Compared with the traditional rolling sphere structure, this 3D fractal strategy realizes the face contact mode and greatly increases the effective contact area. In addition, the rational multi-dimensional fractal strategy and lightweight design enable the sphere to achieve frictional motion when excited by wave energy in any direction of the circumference.

Through the described cutting strategy, two pairs of friction pairs in lateral sliding mode and one pair of friction pairs in contact-separation mode in different directions will be formed between the two component units, realizing the coupled outputs of the two working modes. The working principle of the lateral sliding module is shown in Fig. 2(a), and (i-iv) demonstrates the whole process of the PTFE film and the copper electrode inside the sphere rubbing against each other during the movement and outputting AC power externally through electrostatic induction. Before being excited, the sphere is in a closed state, and the two triboelectric layers are charged with an equal amount of dissimilar signs due to the triboelectrification effect (i). When a wave excitation occurs, the spheres begin to separate from each other, while some of the friction surfaces become separated. The generated potential differences will drive the charges movement between the Cu electrodes via an external circuit (ii). The spheres then continue to separate and the remaining working surfaces continue to sliding friction, the electrons will flow from the lower electrode to the upper electrode to counteract the added electrostatic potential difference, until movement to the maximum separation position (iii). When the two friction layers tend to contact again, the potential difference gradually decreases to zero, and the charges return occurs (iv). The working principle of the contact-separation module is shown in Fig. 2(b), where the same two friction layers are in the starting state with equal amounts of dissimilar charges (i). When excited by a wave, the two friction layers start to separate from each other, and the charge move between the two Cu electrodes driven by a potential difference (ii). The charge movement continues until the maximum separation position (iii). When the two friction layers tend to come into contact again, the potential difference gradually decreases to zero, at which point charge return occurs (iv). Through such periodic contact-separation and lateral sliding, the TENG will generate an AC current pulse signal to output electrical energy to the outside world. In addition, Fig. 2(c) illustrates a comsol simulation showing the trajectories of the two working units under the waves, and the simulation clearly demonstrates the process of potential distribution on the friction surfaces inside the two units. (Video S1, Supporting Information)

Supplementary material related to this article can be found online at

2.2. Dynamical optimization of the FDS-TENG

To further investigate the effect of the cutting fractal strategy on the

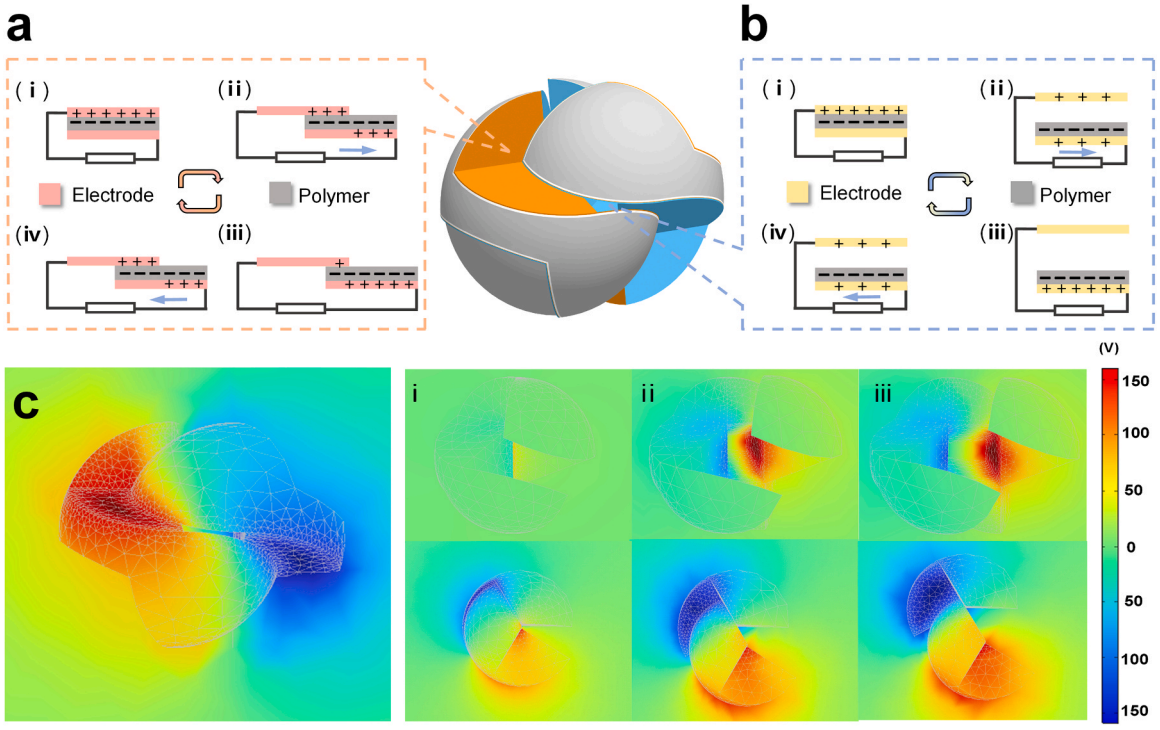


Fig. 2. Schematic of the working mechanism of TENG. a-b) Simplified schematic of the charge distribution and current direction in one cycle in a) i- iv) the lateral sliding part and b) i- iv) the contact-separation part of the TENG. c) The overall potential distribution between the two modules of the TENG and i-iii) the potential distribution under three typical stages.

dynamic and electrical performance of the FDS-TENG, the cutting depth ratio of the sphere was first analyzed, with the structural modelling schematic shown in Fig. S3. The cutting of the sphere in the lateral and vertical directions will generate fan-shaped lateral sliding surfaces and triangular contact-separation friction surfaces, and different ratios of the cutting depth affecting the magnitude of the friction area and the response characteristics under wave excitation, as the given formulas shown in Note. S1. The relationship between the depth of cut ratio and the friction area is shown in Fig. 3(a) for a 3 cm radius sphere. The area of the lateral sliding part gradually increases with the increased depth ratio and reaches 73.3 cm^2 at 75% of the cutting depth, while the friction area of the contact-separation part is 20.3 cm^2 at 25% and gradually decreases to 0 with the decrease of the cutting depth. In this case, the total area tends to decrease slightly with the increase of the cutting depth. Notably, different depth of cutting ratios will not change the coupling of the units but will alter the angle of lateral cutting, thus affecting the response characteristics of the TENG in practice. Here, we introduce the output increase rate at different overturning angles that is the ratio of transferred charge per 5° increase in overturning angle to explore the effect of the cutting depth on the response characteristics. As shown in Fig. 3(b), the 75% cutting depth has superior response characteristics due to having a larger inclination angle of the cutting surface. The sphere completes startup at an overturning angle of 5° and achieves maximum output at 20°. Therefore, with the increase of the cutting depth ratio, the actual utilized area of FDS-TENG is slightly reduced but the response characteristics are further optimized. Considering the above two factors, 75% of the cutting depth ratio is chosen to complete the sphere design. Fig. 3(c) shows the relationship between the friction area and the sphere radius with this cutting strategy and compares it with the surface area of the sphere. Considering the lightweight design

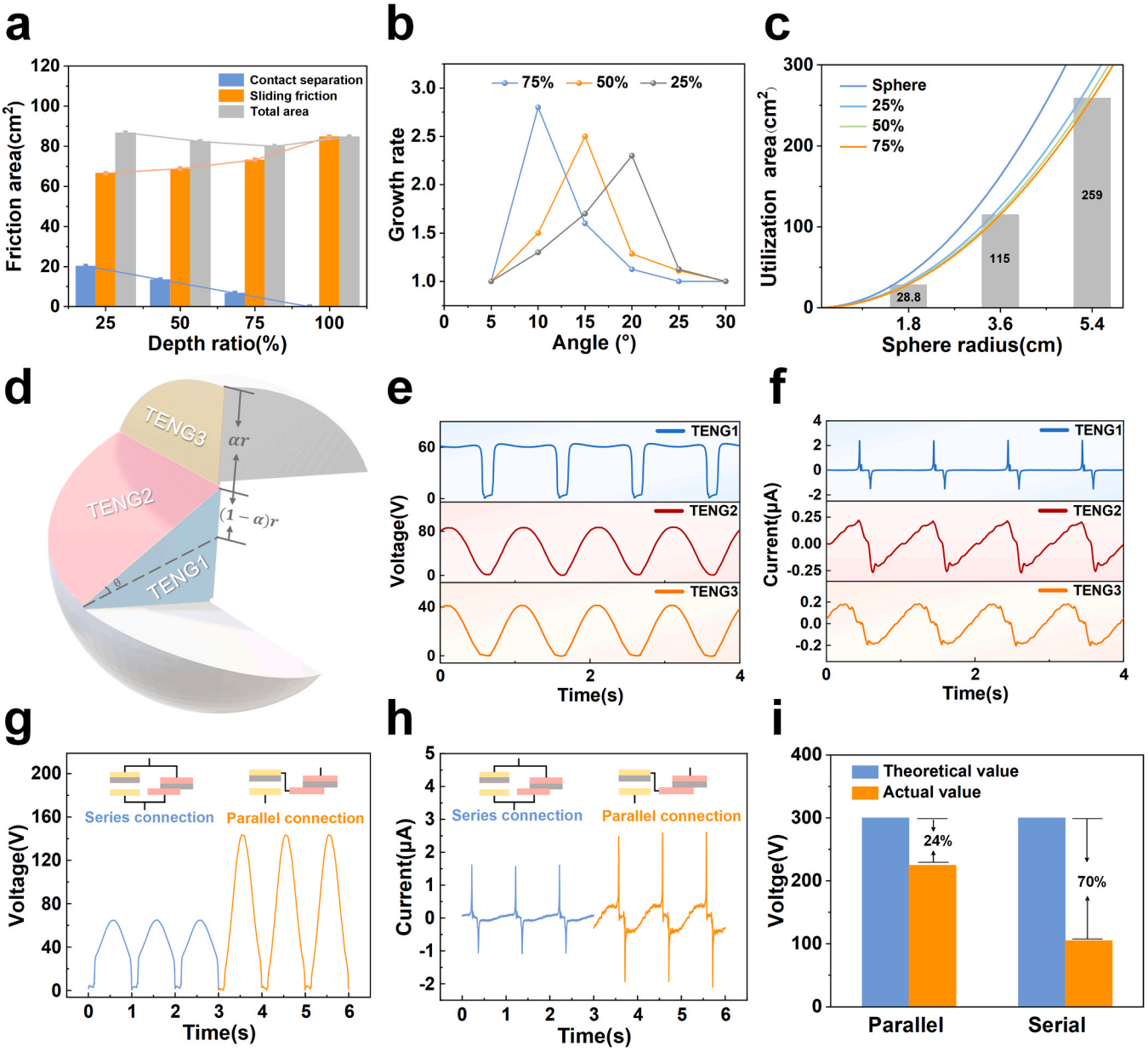


Fig. 3. Structural characterization of the TENG. a) The relationship between the cutting ratio of the TENG sphere and the variation of the friction area. b) The growth rate of the electrical output response of the TENG under different cutting ratios for various overturning angles. c) The comparison of the friction area utilization between the surface area of the sphere and that of the modified structure for different radius. d) Schematic diagrams of each part of the friction unit of a single module of the TENG. e) Open-circuit voltages (V_{oc}) and f) Short-circuit currents (I_{sc}) of the friction layers of the individual units at the frequency of 1 Hz. g-h) Cumulative electrical output effect of the unit friction layer of different circuit connections. i) Comparison of the cumulative effects of the open-circuit voltages.

and portability, the 3.6 cm radius sphere was moderately selected for the subsequent study. The 3.6 cm radius sphere generates a friction area of approximately 115 cm^2, accounting for 71% of the surface area of the sphere, so that the designed sphere structure achieves a greater effective area utilization. The results of the comparative experiments are shown in Fig. S4.

To test the electrical output performance of the FDS-TENG, a swing motor with controllable frequency and tilt angle is used to simulate the wave energy in the lab. Since the single sphere module has three unit surfaces with different characteristics inside, which are classified into TENG1, TENG2, and TENG3, as shown in Fig. 3(d). The basic electrical output characteristics of the three units were tested under the working condition of 1 Hz frequency and 20° tilt angle. Fig. 3(e) demonstrates the short-circuit current (I_{sc}) curve of the unit face, with a value of 2.4 μA for TENG1, and about 0.2 μA for TENG2 and TENG3. Fig. 3(f) illustrates the open-circuit voltage (V_{oc}) curve of the units, with a value of 60 V for TENG1, 85 V for TENG2, and 42 V for TENG3. The amount of charges transferred by the units is shown in Fig. S5. During the complete operation, each lateral sliding TENG generates in-phase electrical signals due to the synchronized and co-coupled motion of the individual components. Also, the relative positions of the contact-separated TENG and the lateral sliding TENG result in their maximum outputs being approximately in the same phase. Since the complexity of the three-dimensional configuration will result in a more cumbersome external circuit connection, the FDS-TENG internal connections were investigated to simplify the back-end circuit. Three neighboring unit-planes were investigated in series and parallel modes as shown in Fig. 3(g, h), with the V_{oc} value of 65 V and the I_{sc} value of 1.6 μA in series mode, and the V_{oc} value of 150 V and the I_{sc} value of 2.7 μA in parallel mode. The amount of charge transferred by the two modes of connection is shown in Fig. S6. Obviously, the waveforms and amplitudes of the unit-TENG outputs in parallel mode can be effectively superimposed. Fig. 3(i) compares the theoretical values with the magnitudes of different connection modes, it is found that the parallel mode reduces the

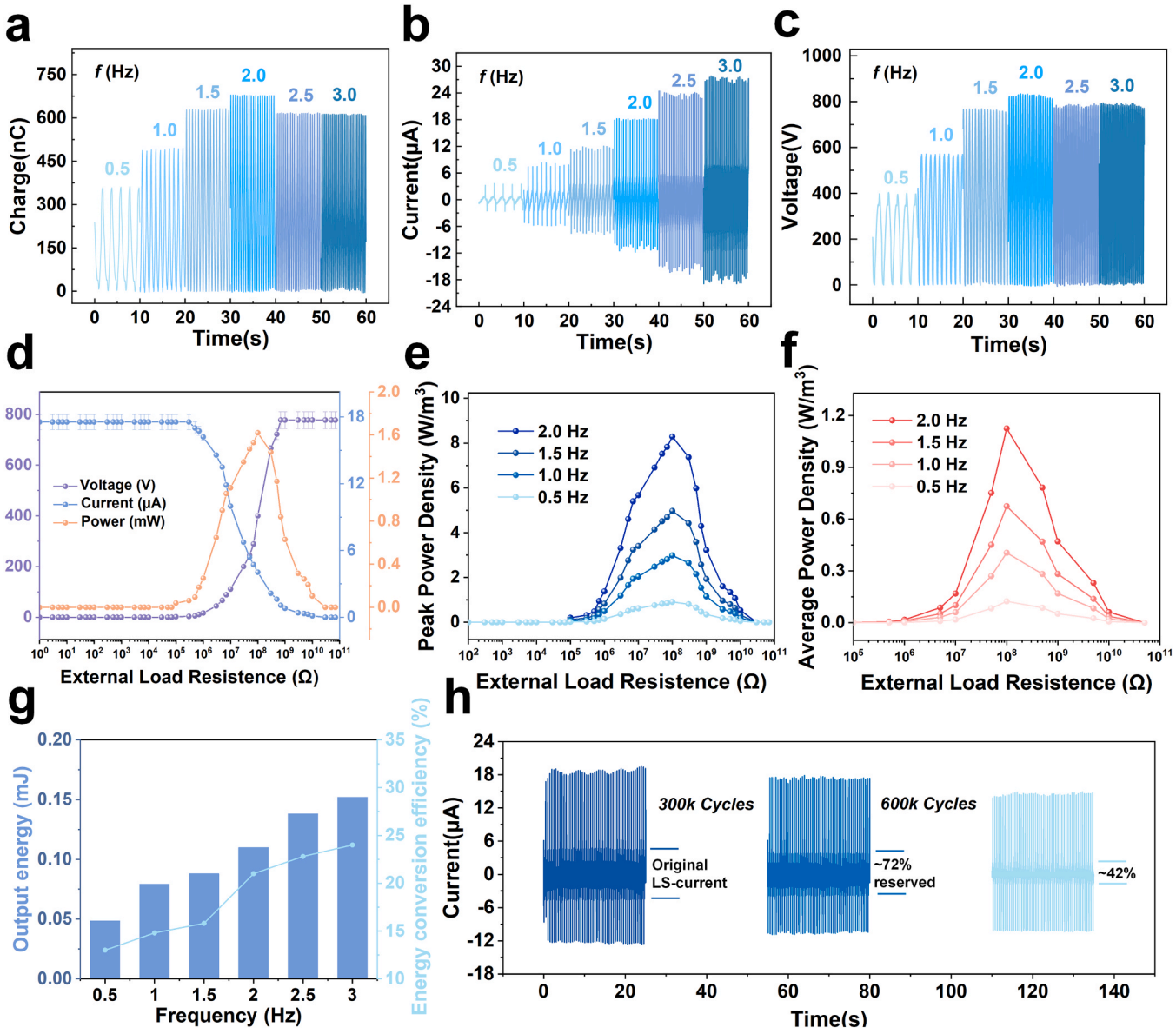


Fig. 4. The electrical output performance of the TENG. a) Transfer charge quantity (Q_{sc}), b) I_{sc}, and c) V_{oc} of the TENG under several swing frequencies controlled by the swing motor. d) Matching impedance measurement of TENG. e) Output peak power and f) output average power with external load resistance at different working frequency. g) Output energy and energy conversion efficiency corresponding to the working frequency. h) Comparison of the output current of an initially fabricated TENG with that of a TENG operated for several cycles.

magnitude by approximately 24% compared to the ideal case, which can be attributed to the effect of parasitic capacitance generated by the neighboring TENGs and the charge cancellation between the multiple TENGs. Based on the above study, we identified the parallel mode to complete the circuit connection inside the sphere. The electrical output performance of all units between the two sub-modules of FDS-TENG is shown in Fig. S7. In addition, the dynamics of the FDS-TENG are discussed in Fig. S8, in which the contact-separation unit can produce a full electrical output at the beginning of the FDS-TENG activation, while the electrical output of the lateral sliding unit increases further with the displacement distance of the coupled motion. Therefore, in high frequency and low wave height wave environments, the area of the contact-separation unit can be optimized to increase with a larger peak current output. On the other hand, in the low frequency and high wave height wave environments that drive the FDS-TENG to overturn more efficiently, it is possible to focus on increasing the area of the lateral sliding unit to achieve higher energy conversion efficiencies throughout the operating cycle.

2.3. Output performance of the FDS-TENG

The integrated electrical output performance of the FDS-TENG at different frequencies was tested by fixing the swing angle to 20°, and the tested curves of the V_{oc}, I_{sc}, and Q_{sc} are shown in Fig. 4(a-c). The values of V_{oc} and Q_{sc} gradually increase to a maximum value as the frequency increases from 0.5 Hz to 2 Hz, and then decrease and stabilize as the frequency continues to increase to 3 Hz. Hence, the maximum values of V_{oc} and Q_{sc} are reached at a frequency of 2 Hz, which are 820 V and 680 nC, respectively. This trend is attributed to the fact that the motion of the TENG at low-frequency band does not enable it to complete the full course of motion until the frequency reaches 2 Hz as shown in Fig. S9, whereas V_{oc} and Q_{sc} gradually decrease after 2 Hz due to the reduction of friction charge caused by the insufficiency of contact during this transient process. The peak value of I_{sc} shows an increasing trend with the frequency, and reaches a maximum value of 26 μA at 3 Hz. Fig. 4(d) shows the output voltage, output current and output power of the FDS-TENG connected to the load resistance at 2 Hz. The amplitude of the output current decreases as the load impedance increases, while that of the output voltage shows an opposite trend. Fig. 4(e, f) shows the peak output power density(E_p)and average output power density (E_a) of the FDS-TENG at different wave frequencies. The E_p and the E_a are calculated by the following equations, where T is the period of operation, U is the load voltage, I is the load current, and v is the sphere volume.

$$E_p = \frac{I_{max}^2 R}{v} \tag{1}$$

$$E_a = \frac{\int_0^T UIdt}{Tv} \tag{2}$$

both of which are positively correlated with the wave frequency and show the trend of increasing followed by decreasing with the increase of the impedance, corresponding to a maximum peak output power density of 8.3 W/m^3 and average power density of 1.1 W/m^3 at 2 Hz with a matched load resistance. As demonstrated in Fig. 4(g), the output energy significantly increases from 0.05 mJ to 0.16 mJ in one cycle with increasing frequency. The electromechanical conversion efficiency (η) of the FDS-TENG at various frequencies is calculated using the following equation, where m is the weight of the moving unit, g is the gravitational acceleration, l is the sliding distance, α is the tilting angle, and μ is the friction coefficient.

$$\eta = \frac{E_{out}}{E_{in}} \cdot 100\% = \frac{\int_0^T I^2 R dt}{mg/\sin\alpha + \mu mg/\cos\alpha} \cdot 100\% \tag{3}$$

In addition, the excellent durability of the contact-separation parts and the mutual support of the coupled three components further slow down the overall wear process. The sustainability of the FDS-TENG was confirmed by a long-term evaluation as shown in Fig. 4(h). Although the larger friction loss in the lateral sliding part causes the current in the sliding part to decay to 42% after 600 k cycles, the overall short-circuit current of the TENG remains relatively stable after running for 300 k and 600 k cycles, reaching 94% and 83% of the electrical output, respectively. The changes in open-circuit voltage and transferred charge are shown in Fig. S10.

Besides the frequency response, the overturning angle of the device caused by different wave heights is an important influence on the electrical performance of the FDS-TENG due to its structural characteristics and kinematic mechanism. Fig. 5(a) illustrates the actual trajectory of the FDS-TENG with wave excitation, where the FDS-TENG undergoes tilting by forced vibration and inertial force thus achieving the open-close operating condition. In order to investigate the ability of FDS-TENG to capture wave energy under different sea conditions, the effects of different overturning angles and swing frequencies on the electrical output of TENG are studied. As shown in Fig. 5(b, c), the FDS-TENG achieves significant electrical output at different swing frequencies (0.5–2.5 Hz) and different overturning angles (10–25°), which demonstrates the ability of the TENG to capture energy effectively under both weakly-excited and strongly-excited wave conditions. In the frequency range of 0.5–2.5 Hz, the value of V_{oc} tends to increase with the increase of the overturning angle, which is attributed to the fact that the TENG will be activated faster and perform coupled motions with larger friction area under greater external excitation and gravity. However, with a further increase in overturning angle and frequency, the fast coupling motion of the TENG will lead to a rapid separation of the PTFE and copper films before full contact resulting in a drop in output voltage. Similarly, because of the positive correlation between the value of I_{sc} and the operating frequency, the output current of the TENG gradually rises but the growth rate decreases as the frequency and the overturning angle increase. At an overturning angle above 20°, the FDS-TENG activates easily and a high voltage output is generated above 400 V over a frequency range of 0.5–2.5 Hz. Under weak excitation conditions with frequency below 1 Hz and overturning angle below 20°, the FDS-TENG is not fully operational, but the internal contact-separation units still provide an output voltage of more than 200 V. Owing to the structural design and dual mode output of the FDS-TENG, the overturning angle of the device will compensate for the low-frequency defects, enabling the TENG to maintain a high electrical output over a wide frequency range of 0.5–2.5 Hz.

Further, the output performance of FDS-TENG under multi-directional wave excitations in the horizontal circumferential range is explored to cope with the randomness of waves in real sea conditions. Firstly, a fixed acrylic frame was installed on the swing motor, and the TENG sphere was mounted on the acrylic frame at different initial angles. Then the frequency of the swing motor was fixed at 2 Hz with a wider range of output variation within 5–20° overturning angle and the deflection angle was set at 30° intervals within a 360° circumference. The variation of transfer charge with overturning angle and deflection angle is shown in Fig. 5(d) and the variation of open-circuit voltage and short-circuit current is shown in Fig. S11. At an overturning angle of 5°, the TENG generates electrical output only within ±30° of the initial 0° angle, that is the sliding direction of the TENG sphere components. As the overturning angle increases which means larger external excitation, when the wave direction is contrary to the sliding direction of the TENG, the sphere will rotate along the excitation direction due to the inertial force of the wave excitation and the gravity of the sphere components, which is attributed to the three-module multi-directional coupling strategy designed for the sphere structure. At an overturning angle of 10°, the TENG produces a similar response in any direction of the circumference, and when the overturning angle is further increased to 20°, the TENG achieves a coupled motion with a maximum sliding distance in any direction of the circumference as well as generating an optimal electrical output.

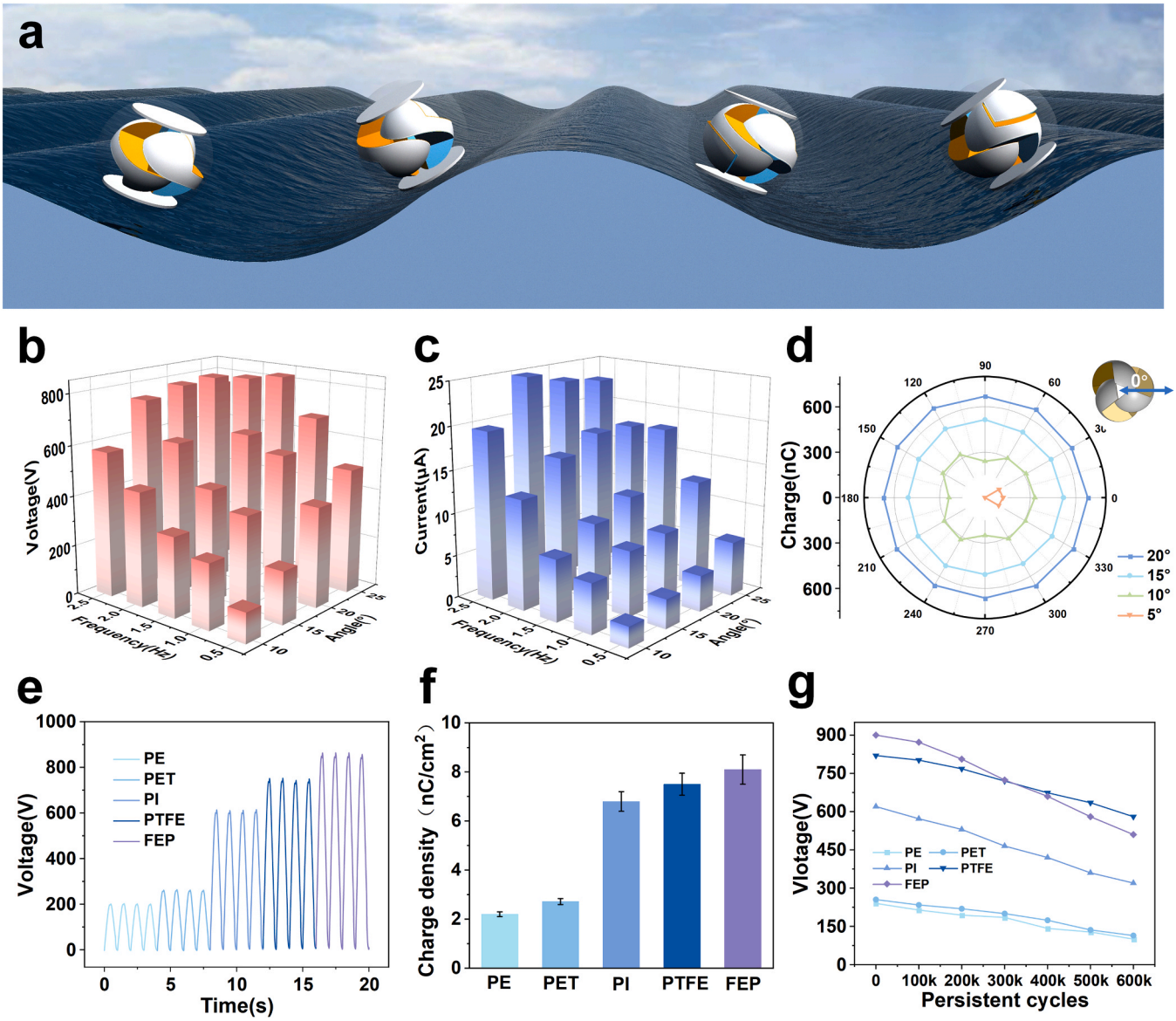


Fig. 5. Influence of structure on output performance of TENG. a) Schematic scenes of various states of motion of the TENG under wave conditions.3D histogram of b) V_{oc}, and c) I_{sc} on changing both the working frequency and swing amplitude. d) Radar diagram of the electrical output performance of TENG for different directions of motion within 360° of horizontal circumference. e) V_{oc}, f) surface charge density, and g) Durability of V_{oc} values for the TENG fabricated with different triboelectric materials.

Since the motion mechanism of TENG mainly relies on the sliding separation of the sphere components, multiple factors such as the friction coefficient, electronegativity, and durability of the friction layer materials will jointly affect the dynamic and electrical output performance of TENG. The common available friction materials such as polyethylene (PE), polyethylene terephthalate (PET), polyimide (PI), fluorinated ethylene propylene copolymer (FEP), and PTFE were selected with thicknesses of 100 μm to explore the output performance of the different materials as friction layers with the copper electrodes of the FDS-TENG. As shown in Fig. 5(e, f), it is observed that the FEP film has the best electrical output performance with an open-circuit voltage of 850 V and a charge density of 8 nC/cm^2 under the optimum operating conditions with frequency of 2 Hz and overturning angle of 20°, which can be attributed to the stronger electronegativity and electron affinity of FEP. On the other hand, the friction coefficients of different materials were tested by tensiometer as shown in Fig. S12, where PTFE has the lowest friction coefficient. Higher friction coefficients will hinder the effective movement of TENG under weak excitation and increases sliding wear, thus reducing the energy conversion efficiency. In addition, the values of V_{oc} for the long-term cycle test can reflect the durability of different materials as shown in Fig. 5(g). The values of V_{oc} for PE, PET, PI, FEP, and PTFE decreased by 58%, 53%, 48%, 43%, and 30%, respectively, after 600 k cycles, while the V_{oc} value of FEP will be lower than that of PTFE. Taken together, the above results show that PTFE combines strong electrical output effect and durability, so that it is an effective and stable material to be used as the friction layer material for the FDS-TENG.

3. Demonstration of harvesting low-frequency wave energy

In order to demonstrate the potential application of FDS-TENG in real marine environments, a distributed sensor self-powered system based on wave energy harvesting is constructed as shown in Fig. 6(a). Several FDS-TENGs composed of a network are connected to the common electronics through energy management circuits. With the excitation of wave energy, the TENG network converts the mechanical energy

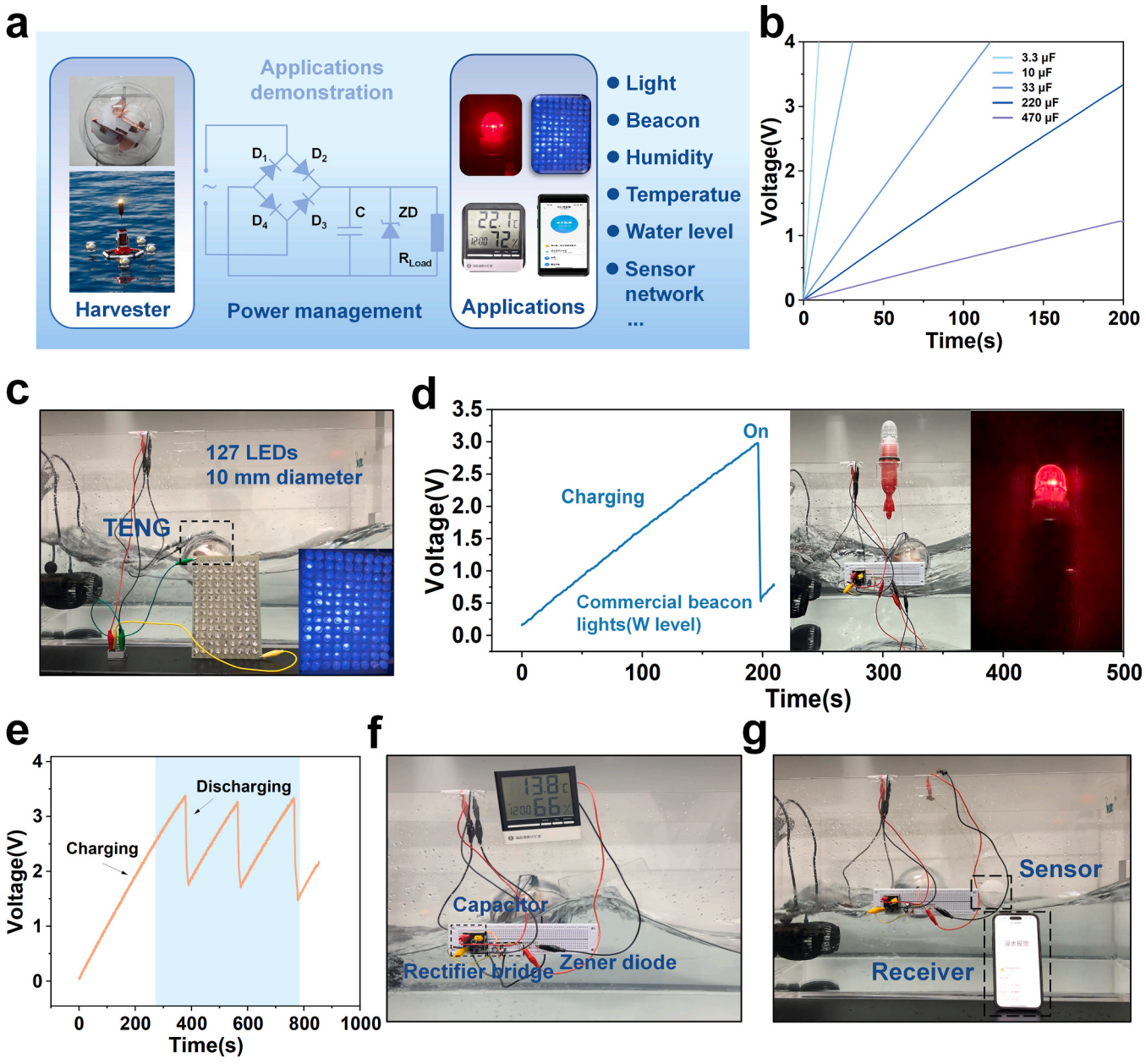


Fig. 6. Application demonstration of FDS-TENG. a) The sensor self-powered system based on FDS-TENG. b) Charging performance of FDS-TENG at 2 Hz. c) Schematic and charging curves of powering a commercial beacon light. d) The voltage curve for the FDS-TENG to power a temperature/humidity meter. e) FDS-TENG driving 127 arrays of 10 mm LEDs. Actual image of f) a temperature/humidity monitoring and g) water level warning with wireless signal transmission.

into electrical energy to provide energy supply for the terminal devices. In the lab, we simulate the wave excitation in realistic environments by generating water waves with specific frequencies and intensities through a pool and a wavemaker, with the dynamic parameters shown in Fig. S13. Fig. 6(b) demonstrates the ability of FDS-TENG to charge different sizes of capacitors at 2 Hz, with 3.3 μF, 10 μF, 33 μF, and 220 μF capacitors charging up to 3 V in 12, 25, 84, and 160 s, respectively. Further, focusing on the practical requirement of the smart oceans, we demonstrate four kinds of devices, including LED arrays, a navigational light for direction guiding, a temperature/humidity meter, and a water level detector with Bluetooth transmission function. As shown in Fig. 6 (c), the FDS-TENG can effectively drive an array of 127 units of 10 mm LEDs for continuous operation with wave excitation (Video S2, Supporting Information). In the actual marine environment, influenced by the complex weather, the light intensity and visible distance of the beacon light impose higher requirements, so that we adopt a 220 μF capacitor for power storage and charging of a commercial beacon light with a power of 3 W. Fig. 6(d) shows the effect of charging a commercial beacon light and the capacitor charging curve. After 180 s of continuous charging, the capacitor will be charged to the operating voltage for the beacon light of 3.2 V, at which time the beacon light will be activated and emit strong light (Video S3, Supporting Information). In practical applications, more efficient operation of the beacon light will be achieved by TENG array network and power management. In addition, Fig. 6(e) shows the charging curve of the FDS-TENG successfully driving the operation of a temperature/humidity meter with an actual diagram shown in Fig. 6(f). A 330 μF capacitor was used to store electrical energy to power the temperature/humidity meter, when the capacitor was charged to 3.4 V and the switch was turned on, the meter started to work and display the ambient temperature and humidity, and then entered into the intermittent operation mode (Video S4, Supporting Information). Finally, we tried to use FDS-TENG to jointly drive a water level

alarm with Bluetooth transmission. Here, a 2 mF capacitor was charged to 3 V to power the water level alarm. When the water level exceeds a pre-set threshold, the water level alarm warns and wirelessly transmits to a mobile phone application via Bluetooth as shown in Fig. 6(g) and (Video S5, Supporting Information). The above applications show that the FDS-TENG for wave energy collection can effectively meet the energy supply needs of intelligent ocean sensors, which is of significant application value to enhance the endurance of ocean monitoring equipment.

Supplementary material related to this article can be found online at

4. Conclusion

In summary, a dual-mode spherical TENG based on an efficient fractal strategy is proposed for harvesting low-frequency wave energy. A three-module multiphase coupling working principle is introduced through 3D structural design, whose structure allow the various components of the FDS-TENG to be coupled with each other and achieve efficient synergistic motion after optimization. Meanwhile, the proposed multi-dimensional cutting strategy greatly increases the actual utilized friction area of the spherical TENG and forms a dual-mode output with contact-separation mode and lateral sliding mode, where the different modes of operation are able to effectively complement each other in the random, multi-directional and weak wave excitation to maintain a satisfactory energy conversion efficiency of the TENG in the overall frequency range of wave energy. Compared with the traditional rolling sphere structure, the FDS-TENG retains similar dynamic characteristics but outputs better electrical output performance in low-frequency range. The experimental results confirmed that FDS-TENG can effectively respond to low-frequency wave excitation in the range of 0.5–3 Hz. At an operating frequency of 2 Hz and an overturning angle of 20°, the FDS-TENG has a maximum peak power density of 8.3 W/m^3 and an average power density of 1.1 W/m^3. The durability and stability of the FDS-TENG is also verified over a long period of 600 k cycles, with a 17% decrease in short circuit current for PTFE. Finally, the FDS-TENG was demonstrated to light up 127 LEDs and to drive several small electronic devices. The FDS-TENG was successfully demonstrated in a pool driving commercial beacon lights, temperature and humidity meters, and water level alarms with Bluetooth wireless transmission, confirming that TENG has great potential for marine IoT, environmental awareness, and safety monitoring.

5. Material and methods

5.1. Fabrication of the FDS-TENG

The FDS-TENG mainly consists of two components: three identical TENG sphere components and acrylic encapsulated shell. For TENG, the model structure was constructed by the modelling software Solidworks with a regular sphere as the base model (3.6 cm radius). By means of 3D printing, three identical hollow sphere components as described were fabricated (in resin material with a thickness of 2 mm and a weight of 24 g). Copper foil (50 μm thickness) was applied to five working surfaces on the one side of the sphere as electrodes, and a sponge (1 mm) and a copper-coated PTFE film (100 μm thickness) were sequentially applied to five working surfaces on the other side of the sphere as a friction layer. In order to enhance the surface roughness and triboelectrification, the PTFE films were placed in an inductively coupled plasma instrument (STS LPX ASE-SR) chamber and etched by 400 W, 40 kHz air plasma for 10 min. After that, the treated samples were ultrasonically cleaned with deionized water and ethanol successively, then dried in a drying oven before use. The above three identical sphere components are cross-coupled with each other on opposite sides to form a sphere structure TENG that can move in concert. For the acrylic encapsulation shell, A round acrylic shell (radius 6 cm, thickness 2 mm) was used as the encapsulation shell, and circular fixing plates were mounted 2 cm apart from the top and bottom ends of the shell. In the center of the upper and lower fixed plate connected to the upper and lower ends of a TENG sphere component through a rotating bearing, forming a rotating structure with this as the central axis.

6. Experimental measurements

To evaluate the output performance of the TENG, the TENG was mounted on a swing motor to simulate wave excitation at different frequencies and different wave heights. The electrical signals of the TENG were measured by a programmable electrostatic meter (Keithley 6514). For multi-channel measurements, the voltage signals were measured by an ADC (NI 9220, 16-channel voltage measurement module, ± 10 V). The measured signals were visualized in a high-speed data acquisition system under LabView 2016. The performance testing and application demonstration of TENG was carried out in a conventional pool with water waves generated by a wave-making device (RW-20).

CRediT authorship contribution statement

Hao Luo: Visualization, Validation, Formal analysis. **Zhentan Quan:** Visualization, Software, Methodology. **Lijun Yang:** Visualization, Validation, Resources. **Ruijin Liao:** Supervision, Project administration. **Boyu Zhu:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Jiyu Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Hanqing Wang:** Methodology, Conceptualization. **Han Wu:** Validation, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2024.109446.

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